Healthcare Agent for Drug Molecule Generation

Team – 04

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1) Architecture Workflow:

- **User Input -**
  The user provides a query or request for generating drug-like molecules.

- **Retrieval Module (RAG) -**
  The query is converted into an embedding and similar molecules are retrieved from the knowledge base.

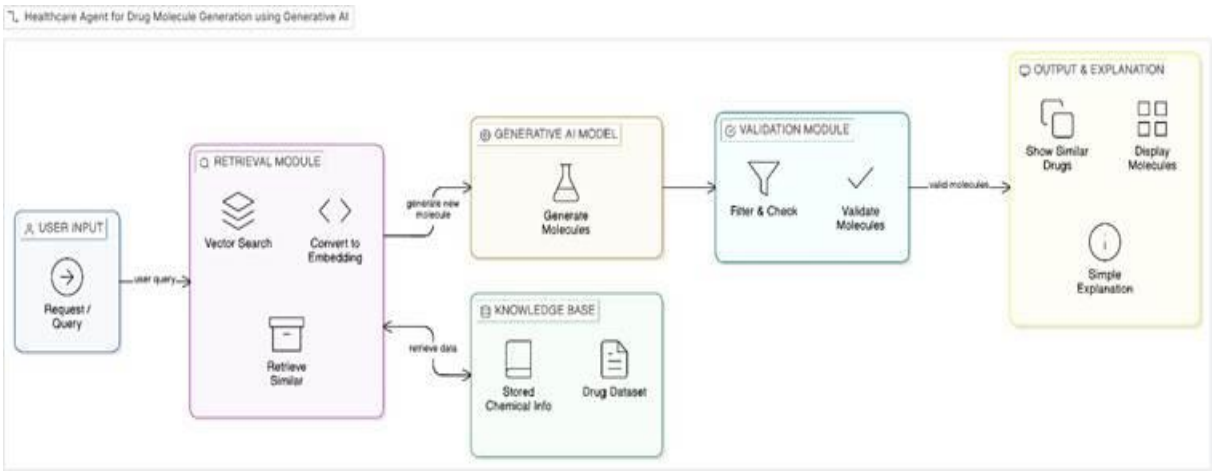
- **Generative AI Model -**
  The trained model generates new SMILES-based molecular structures using learned patterns and retrieved context.

- **Validation Module -**
  Generated molecules are validated using RDKit to ensure chemical correctness. Invalid molecules are removed.

- **Output & Explanation -**
  Valid molecules are displayed along with similar known drugs for reference.

Architecture Diagram:

User Query → Retrieval → Generation → Validation → Output



2) Experiment Design:

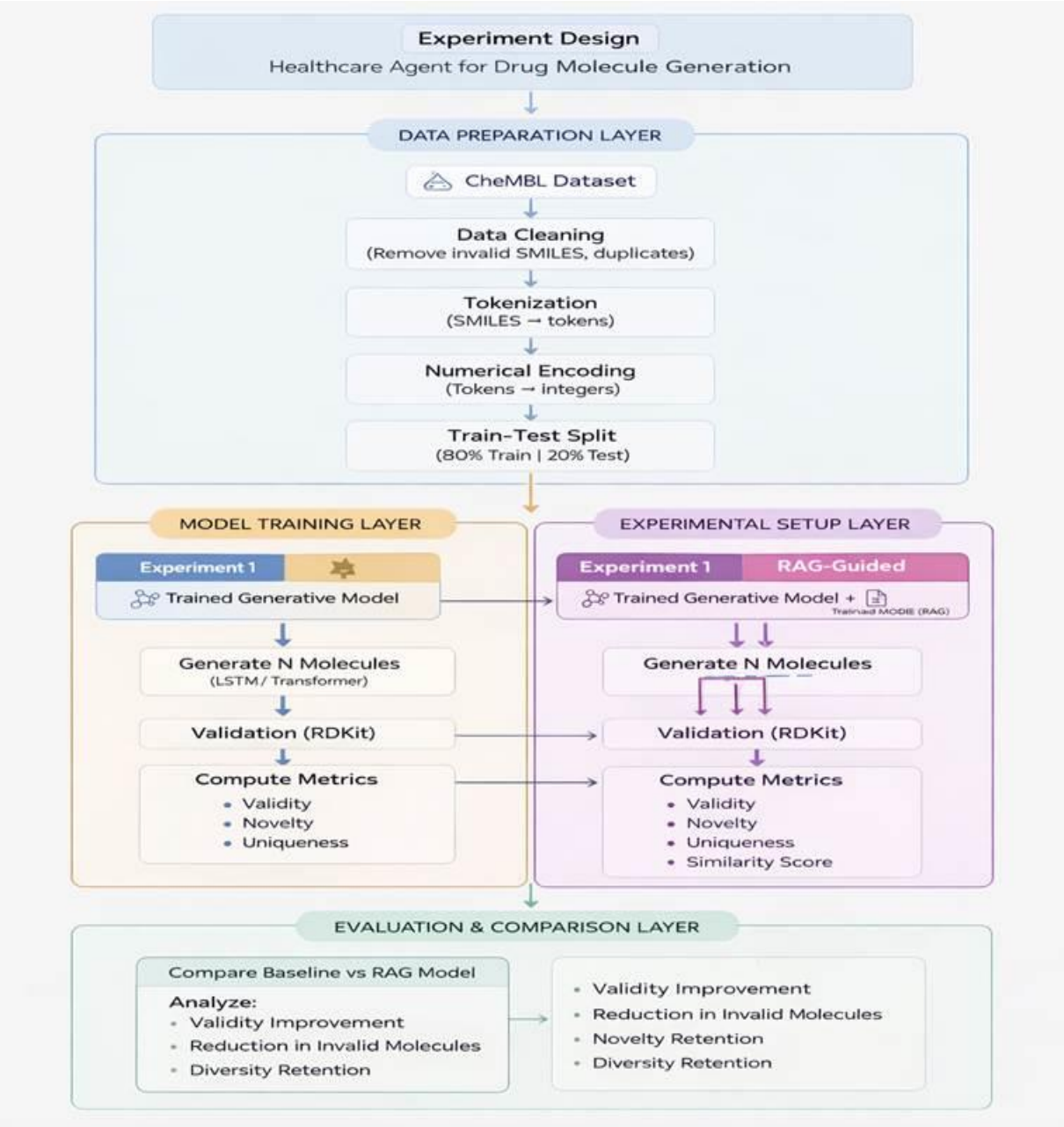
- Data Preparation
  - Use ChEMBL dataset
  - Clean data and remove invalid SMILES
  - Tokenize and encode
  - Split into training and testing sets

- Model Training
  - Train the generative model using the training dataset.

- Experiment 1 – Baseline
  - Generate N molecules
  - Validate using RDKit
  - Compute Validity, Novelty, and Uniqueness

- Experiment 2 – RAG-Guided
  - Retrieve similar molecules
  - Generate N molecules
  - Validate using RDKit
  - Compute same metrics

- Evaluation & Comparison
  - Compare baseline and RAG results to analyze improvement in molecule validity and quality.

Dataset → Train Model → Baseline vs RAG → Validate → Compare Metrics

Experiment Design Diagram: